



Published in final edited form as:

J Allergy Clin Immunol Pract. 2024 April ; 12(4): 870–879. doi:10.1016/j.jaip.2024.01.011.

The Role of ICS-containing Rescue Therapy versus SABA Alone in Asthma Management Today

James G. Krings, M.D., M.Sc.¹, Richard Beasley, D.Sc.²

¹Division of Pulmonary and Critical Care Medicine, Washington University in St. Louis School of Medicine, St. Louis, MO, USA

²Medical Research Institute of New Zealand, Wellington, New Zealand; Victoria University of Wellington, Wellington, New Zealand.

Abstract

The Global Initiative of Asthma (GINA) recommends that short-acting beta₂-agonist (SABA) monotherapy should no longer be prescribed, and that as-needed combination inhaled corticosteroid (ICS)-formoterol is the preferred reliever therapy in adults and adolescents with mild asthma. These recommendations are based on the risks of SABA monotherapy, the evidence that ICS-formoterol reliever therapy markedly decreases the occurrence of severe asthma exacerbations when compared to SABA reliever therapy alone, and because ICS-formoterol reliever therapy has a favourable risk/benefit profile when compared with maintenance ICS plus SABA reliever therapy. Data supporting the use of combination ICS-albuterol reliever therapy in mild asthma are more limited, but there are studies that inform its use in this population. In this review, we compare, using a pros and cons format, the: (1) long-term safety and efficacy of ICS-formoterol reliever therapy vs. SABA reliever therapy alone, (2) long-term safety and efficacy of ICS-albuterol reliever therapy vs. SABA reliever therapy alone, (3) immediate bronchodilator effects of ICS-formoterol vs. SABA alone, and (4) clinical and regulatory factors that may inform reliever therapy prescription decisions. By presenting the evidence of these reliever inhaler options, we hope to inform the reader, while also calling for necessary future effectiveness and implementation research.

INTRODUCTION:

The Global Initiative for Asthma's (GINA) recommendation that short-acting beta₂-agonist (SABA) reliever therapy should not be used alone in the treatment of asthma, represents one of the most fundamental paradigm shifts in the management of asthma over recent decades.¹ This recommendation recognizes that, while patient self-administration of SABAs

Corresponding Author: James Krings, M.D., M.Sc., Washington University in St. Louis School of Medicine, Campus Box 8052, 660 S. Euclid Ave., St. Louis, MO 63110, United States of America, Phone: +1.314.273.4734, kringsj@wustl.edu.

Author contributions:

J.G.K. and R.B. together conceived of and wrote this article in its entirety.

Publisher's Disclaimer: This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

is a simple and effective method to quickly relieve asthma symptoms, SABA therapy alone is associated with substantive risks, and that there are now other simple, and more effective, reliever inhaler options that markedly reduce risk and improve patient outcomes.^{2–8}

The risk of SABAs can be broadly grouped into two groups of adverse effects; those relating to their long-term use and those associated with their overuse in the situation of a severe attack.⁹ Regular use of a SABA can result in beta₂-receptor downregulation, which leads to a reduced magnitude and duration of bronchodilation with future SABA doses, as well as increased bronchial responsiveness.^{10–13} As a result, regular SABA use can make patients more vulnerable to future asthmatic stimuli, and increase the proportion of patients within a community who will experience a moderate or severe asthma exacerbation.¹² In a severe attack, the overuse of SABAs can lead to adverse cardiovascular effects, hypokalemia, hypoxaemia, and a delay in seeking medical help, which itself is an important risk factor for a fatal outcome.^{9,14,15} The importance of these risks is highlighted by the mortality epidemics that occurred following the introduction of isoprenaline forte and fenoterol, which were two high-dose preparations of potent and poorly selective beta₂-agonists.^{9,16} Furthermore, more gradual increases in asthma mortality occurred in many countries during the 1960s to 1990s, associated with increasing use of beta₂-agonists as an entire class.^{16–18} Finally, the Salmeterol Multicenter Asthma Research Trial demonstrated that regular long-acting beta-agonist (LABA) use, when not paired with a concomitant inhaled corticosteroid (ICS), increased asthma mortality risk with findings that were particularly concerning amongst African American patients.¹⁹ In contrast, there was no increased risk of serious asthma-related events (death, endotracheal intubation, or hospitalization) with combination ICS-LABA use compared with ICS alone.²⁰

The prior morbidity and mortality concerns associated with SABAs were partially mitigated by selecting ‘safer’ SABAs, such as albuterol, and changing SABA prescriptions from ‘regular 4x daily plus as-required’ to ‘as-required’ only. However, these measures did not fully mitigate the risks associated with SABAs. ‘As-required use’ of SABAs essentially results in repeated SABA use when a patient has poor symptom control, which exposes patients to both the long- and short-term risks associated with SABAs. In stark contrast to SABAs, long-term use of ICSs improves lung function, reduces the risk of severe asthma exacerbations, and decreases asthma-related mortality.^{21–24} The beneficial effects of ICS therapy occur even in patients with so-called ‘intermittent asthma’, and in patients prescribed ‘low doses’ of ICSs.^{21,25,26} While traditionally patients with mild asthma are prescribed maintenance ICS inhalers to gain the beneficial effects of ICS exposure, poor maintenance inhaler adherence represents a common and intractable problem.^{27–30}

Recognizing the major limitations of the traditional treatment paradigms in mild asthma, asthma researchers have investigated the combination of an ICS with either a SABA or a fast-onset LABA (specifically formoterol) within a single inhaler device to be used as a reliever.^{31–33} This therapeutic approach ensures ICS delivery during symptomatic periods when a patient uses their reliever but does not take their maintenance ICS-based treatment, and also enables titration of ICS dosing, through the vehicle of reliever inhaler usage. An essential requirement for the reliever SABA or LABA used in combination with an ICS is that it is fast acting, thereby providing patients quick symptom relief. To date this approach,

which has been termed ‘anti-inflammatory reliever therapy’, has been investigated with ICS-albuterol and ICS-formoterol, with evidence that both approaches are more effective, and safer, than SABA reliever therapy alone across the range of asthma severities.⁶

Despite the scientific evidence and GINA recommendations, ICS-containing reliever therapy is significantly under-prescribed and under-utilized across the world.^{34–39} The reasons for this ‘implementation gap’ (i.e., the difference between an evidence-based recommendation and real-world adoption) in inhaler prescription practices are complex and include entrenched clinical practice, high inhaler cost, regulatory obstacles, and other barriers.^{6,8,39}

In this article, we focus on the use of ICS-containing reliever therapy alone in comparison with SABA reliever therapy alone in mild asthma and discuss the pros and cons of these reliever options. Moreover, we emphasize the benefits and shortcomings to ICS-containing reliever therapy, gaps in the underlying data, and barriers to real-world implementation (Table 1).

1. ICS-FORMOTEROL RELIEVER THERAPY VS. SABA RELIEVER THERAPY IN MILD ASTHMA: COMPARISON OF EFFICACY AND SAFETY WITH LONG-TERM USE:

Pros of ICS-formoterol reliever therapy alone:

- i. *Severe exacerbations:* ICS-formoterol reliever therapy reduces the number of severe exacerbations requiring systemic corticosteroids vs. SABA reliever therapy alone by approximately half (Table 2).⁴⁰ Moreover, vs. SABA reliever therapy alone, ICS-formoterol reliever therapy reduces the occurrence of asthma-associated hospital admissions, emergency department (ED) visits, or urgent care visits by approximately two-thirds (Table 2).⁴⁰

The *absolute* reduction in exacerbation risk associated with reliever ICS-formoterol vs. reliever SABA therapy alone in mild asthma is similar to the absolute risk reduction associated with reliever ICS-formoterol vs. reliever SABA in patients with moderate-to-severe asthma also taking maintenance ICS-LABA therapy, while the *relative* risk reduction with ICS-formoterol vs. SABA alone in mild asthma is even greater (Figure 1).^{40,41}

As hospital admissions can be considered a surrogate for mortality risk,⁴² these findings suggest that ICS-formoterol reliever therapy may reduce asthma-associated mortality compared with SABA reliever therapy alone. Notably, reliever ICS-formoterol therapy reduces the occurrence of severe exacerbations, compared with SABA therapy, even in patients with so-called ‘intermittent asthma’, who use SABAs ≤ 2 times per week.⁴³ In a network meta-analysis, ICS-formoterol reliever therapy alone ranked higher than SABA reliever taken together with low-dose maintenance ICS-LABA, or with low-dose maintenance ICS therapy in preventing the risk of severe exacerbation in mild-to-moderate asthma; notably, SABA reliever monotherapy ranked the least effective of the therapeutic regimens analyzed (Figure 2).⁴⁴

- ii. *Asthma symptoms:* ICS-formoterol reliever therapy improves asthma symptoms more than SABA reliever therapy alone, based on a greater reduction in the Asthma Control Questionnaire–5 (ACQ-5) of 0.15 (Table 2).⁴⁰ However, this difference is below the minimal clinically important difference (MCID) for ACQ-5 of 0.50,⁴⁵ suggesting there is no clinically important difference in day-to-day asthma symptoms between the two reliever regimens. Notably, when considering prophylactic usage to prevent symptoms, ICS-formoterol has been found to be superior to SABAs at preventing exercise-induced bronchoconstriction.⁴⁶
- iii. *Lung function:* ICS-formoterol reliever therapy improves lung function compared with SABA reliever therapy alone, with a 64mL greater increase in the pre-bronchodilator forced expiratory volume in 1 second (FEV₁) (Table 2).⁴⁰ This difference is below the MCID of FEV₁ of 230mL,⁴⁷ suggesting there is no clinically important difference between the regimens regarding lung function.⁴⁷
- iv. *Airway inflammation:* ICS-formoterol reliever therapy, vs. SABA therapy alone, leads to a 17% reduction in the fractional exhaled nitric oxide (FeNO) (Table 2) thereby demonstrating anti-inflammatory airway effects, even with minimal ICS exposure (mean 107µg/day budesonide).⁴⁸ When comparing budesonide-formoterol reliever therapy to formoterol reliever therapy alone in mild asthma, budesonide-formoterol reliever therapy (mean 163µg/day budesonide) results in a decrease in FeNO of 16 ppb (95% CI 8 to 24).⁴⁹
- v. *Adverse events:* ICS-formoterol reliever therapy reduces the occurrence of adverse events by approximately 18% as compared to SABA therapy alone, primarily due to reduced asthma-related events.⁴⁰ Click or tap here to enter text.

Cons of ICS-formoterol reliever therapy alone:

- i. *A paucity of efficacy and safety data in children:* A major limitation of all ICS-containing reliever inhaler paradigms is that they have been significantly understudied in children.⁴⁰ In mild asthma, there have been no studies of ICS-formoterol reliever therapy alone in patients aged <12 years old. In children 4–11 years old with moderate-to-severe asthma, ultra-low-dose ICS-formoterol SMART therapy has shown benefits of greater magnitude than those seen in adolescent and adult asthma.⁵⁰ Considering the high rate of severe asthma exacerbations in children, the limited evidence to inform reliever inhaler choices means that many children may be missing out on the potential benefits of ICS-containing reliever regimens.
- ii. *Generalizability of data to different delivery devices and non-budesonide ICS-formoterol formulations:* In the large randomized controlled trials (RCTs), which informed the latest GINA recommendations, the ICS-formoterol product was budesonide-formoterol 200–6µg delivered via a dry powder inhaler (DPI) turbuhaler device.^{48,51–53} The generalizability of these results to different budesonide-formoterol devices, or other non-budesonide formulations, is unknown.

Some patients may have difficulty mastering the turbuhaler device. One option might be to switch a patient to budesonide-formoterol via a pressurized metered dose inhaler device (pMDI); however, this prescription would then represent ‘off label’ use unless regulatory approval for such use had been granted in a particular location. In some countries (such as the United States [US]), the budesonide-formoterol turbuhaler is not even commercially available at all, and any use of a budesonide-formoterol pMDI on a reliever basis is technically ‘off label.’ Nonetheless, it is reassuring that the efficacy and safety data for SMART using budesonide-formoterol were similar between the pMDI and DPI delivery devices.⁵⁴ Finally, every study of ICS-formoterol reliever alone specifically utilized budesonide as the ICS component. Therefore, in locations where other ICS-formoterol combinations, such as fluticasone-formoterol, beclomethasone-formoterol, or mometasone-formoterol, are commercially available, the evidence supporting their reliever usage is indirect.

- iii. *Differences in cumulative ICS exposure:* Budesonide-formoterol reliever use in mild asthma results in a mean daily budesonide exposure of about 75ug, which could potentially result in a very small risk of systemic corticosteroid side effects in a small minority of patients.⁴⁰

Summary: The greater efficacy of ICS-formoterol reliever therapy vs. SABA therapy alone in mild asthma is primarily attributed to a marked reduction in severe exacerbation risk, rather than improvements in day-to-day asthma symptoms or lung function (Table 2). The >50% relative reduction in severe asthma exacerbation risk is clinically important as, when changing from reliever SABA alone to reliever budesonide-formoterol alone, only 18 patients would need to be treated for one year to prevent one severe asthma exacerbation, an estimate which is based on high-certainty evidence.⁴⁰

Overall, ICS-formoterol reliever therapy reduces the occurrence of severe exacerbations, improves asthma symptoms, decreases airflow obstruction, and reduces airways inflammation, with a better adverse event profile than SABA therapy alone. The lack of data for this approach in children represents an area of significant need of study. The evidence for a reliever-only ICS-formoterol approach is derived solely from studies of budesonide-formoterol utilising the turbuhaler device. Other ICS-formoterol formulations should be studied as reliever inhalers alone, at different dose combinations, to determine their respective safety/efficacy profiles.

2. ICS-ALBUTEROL RELIEVER THERAPY VS. SABA RELIEVER THERAPY IN MILD ASTHMA: COMPARISON OF EFFICACY AND SAFETY WITH LONG TERM USE:

Pros of ICS-albuterol therapy alone:

- i. *Severe exacerbations:* There is markedly less data supporting the use of combination ICS-albuterol reliever therapy than with ICS-formoterol. In one six-month study, use of beclomethasone-albuterol reliever therapy reduced the risk of severe exacerbations compared with albuterol reliever therapy alone (0 severe exacerbations per 122 patients vs. 10 severe exacerbations per 118

patients over the 6 month period of the study).⁵⁵ The proportion of patients who experienced at least one moderate or severe exacerbation was significantly less in patients who received beclomethasone-albuterol vs. albuterol reliever therapy alone (4.9% vs. 17.8%; $P=0.002$), but similar when compared to the group that received maintenance beclomethasone plus as-needed albuterol (4.9% vs. 5.7%; $P=0.80$) (Figure 3). The mean number of moderate and severe exacerbations per patient per year was less in the as-needed beclomethasone-albuterol than in the as-needed albuterol group (0.74 vs. 1.63, $P<0.001$).

- ii. *Asthma symptoms:* In the one study, beclomethasone-albuterol compared to albuterol reliever therapy alone, improved some measures of asthma control, such as the number of nocturnal awakenings (0.10 less nocturnal awakenings per night; $P=0.03$), but not day or nighttime asthma symptom scores, or symptom-free days.⁵⁵ The clinical significance of the difference in nocturnal awakenings is uncertain.
- iii. *Lung function:* Beclomethasone-albuterol improved lung function with an increase in the pre-bronchodilator FEV₁ of 120mL when compared to albuterol reliever therapy alone ($P=0.007$).⁵⁵ This difference in FEV₁ is below the MCID of 230mL,⁴⁷ suggesting there may be no clinically important difference in lung function between regimens.
- iv. *Adverse events:* The number of adverse events has not been shown to differ between beclomethasone-albuterol and albuterol therapy alone.⁵⁵

Cons of ICS-albuterol reliever therapy alone:

- i. *Paucity of data on ICS-albuterol therapy in mild asthma:* The main limitation to ICS-albuterol reliever therapy is that there has only been a single 6-month study of this regimen in adults with mild asthma with a total of 240 patients randomised to the two regimens.⁵⁵ No study to date has evaluated the safety and efficacy of combination ICS-albuterol reliever therapy alone in adolescents or children aged <18 years.
- ii. *Generalizability of data to different delivery devices and ICS-albuterol formulations:* The only study to date of ICS-albuterol reliever therapy used beclomethasone-albuterol pMDI.⁵⁵ The generalizability of these findings to other ICS-albuterol products is unknown. Notably, in the US, a fixed-dose combination budesonide-albuterol device was recently approved by the Food and Drug Administration (FDA) to prevent or treat bronchoconstriction in patients ≥ 18 years old regardless of asthma severity.⁵⁶ While budesonide-albuterol demonstrated a decreased risk of severe asthma exacerbations compared with albuterol reliever therapy in moderate-to-severe asthma patients who are also using ICS-based maintenance therapy,⁵⁷ it has not been studied as a sole reliever therapy in patients with mild asthma.
- iii. *Differences in cumulative ICS exposure:* ICS-albuterol reliever therapy leads to a greater ICS exposure than SABA therapy alone, and the magnitude of this difference is dependent on the ICS dose used in the combination product.

However, this increased ICS exposure is balanced by the reduced exposure to systemic corticosteroids from a decreased exacerbation risk, and a decreased ICS exposure compared to those escalated to maintenance ICS plus SABA reliever therapy.

Summary: There is limited data on the safety and efficacy of combination ICS-albuterol inhalers in mild asthma. However, the available evidence suggests greater efficacy of ICS-albuterol vs. albuterol therapy alone at reducing moderate and severe exacerbation risk.⁵⁵ The benefits in asthma symptom control and lung function with reliever ICS-albuterol therapy are of uncertain clinical importance. The limited evidence is based on the beclomethasone-albuterol inhaler, not the recently FDA-approved budesonide-albuterol pMDI,⁵⁶ and applies to adults, but not children or adolescents.

Indirect evidence supporting the use of combination ICS-albuterol in mild asthma is derived from the recent landmark RCT which showed that budesonide-albuterol reliever therapy reduced the risk of severe exacerbations vs. albuterol reliever therapy in moderate-to-severe asthma patients on concomitant ICS-based maintenance therapy.⁵⁷ Further indirect evidence supporting the ICS-albuterol reliever approach in mild asthma comes from an RCT in children investigating the regimen in which patients were instructed to actuate a dose of beclomethasone whenever they used a separate albuterol inhaler.⁵⁸ This study found that this dual separate reliever inhaler regimen reduced the time to first severe exacerbation by 38% (HR 0.62; 95% CI: 0.37 to 1.05; P=0.07) compared with a SABA reliever only regimen, though this result was not statistically significant.⁵⁸ Notably, there were significantly fewer treatment failures in the dual separate reliever inhaler use regimen compared with the SABA reliever alone regimen (8.5% vs. 23%, P=0.024).⁵⁸ Finally, indirect evidence supporting the increased safety of ICS-albuterol vs. albuterol comes from the recent FDA-mandated study in mild-to-moderate asthma, which showed that with frequent (4 times daily) use of reliever therapy (as could occur in a period of unstable asthma), severe exacerbations were 4 to 5 times more frequent with frequent albuterol use compared with high- and low-dose budesonide-albuterol.⁵⁹

There have been no head-to-head comparisons of ICS-formoterol reliever vs. ICS-albuterol reliever therapy.⁶⁰ However, as some studies demonstrated a greater efficacy of formoterol vs. SABAs at reducing severe exacerbation risk,^{61–63} it is plausible that ICS-albuterol reliever therapy alone is less effective than ICS-formoterol.

3. ICS-FORMOTEROL RELIEVER THERAPY VS. SABA RELIEVER THERAPY IN MILD ASTHMA: COMPARISON OF EFFECTS ON ACUTE BRONCHODILATION:

Assessments of the comparative acute bronchodilator effects between ICS-formoterol and SABAs have been limited by the absence of definitive studies determining bronchodilator dose equivalence, and complicated by the differing durations of action and intrinsic activity between these betaagonists.^{64–66} However, when administered in the reliever doses most commonly used in asthma, albuterol (100µg × 2 actuations) represents a higher beta₂-agonist dose than budesonide-formoterol (200–6µg × 1 actuation), leading to a greater beta₂-agonist

effect on airways (e.g., change in FEV₁),⁶⁷ and systemic beta₂ agonist effects (e.g. change in serum potassium and heart rate).⁶⁸

- i. *Comparison of bronchodilator effect of a single reliever inhaler dose:* In patients with stable asthma, two actuations of albuterol 100µg pMDI results in a 97mL (95% CI: 47 to 147) greater change in the FEV₁ when compared to one actuation of budesonide-formoterol 200–6µg, when assessed 2 minutes after administration.⁶⁷ When averaged over repeated assessments for 30 mins, the FEV₁ was 100mL (95% CI: 80 to 120) higher for the albuterol group compared with budesonide-formoterol.⁶⁷ These differences are less than the MCID for FEV₁ of 230mL⁴⁷ and were not associated with differences in measures of breathlessness.⁶⁷ Notably, formoterol has a bronchodilator duration of 12–24 hours, compared with 4–6 hours with albuterol.^{69,70}
- ii. *Comparison of bronchodilator effect of repeated reliever inhaler doses:* When bronchodilator response of repeated actuations of albuterol vs. budesonide-formoterol in a 200µg vs. 200–6µg ratio were compared, albuterol resulted in a greater improvement in FEV₁ over the first 4 hours; however, budesonide-formoterol resulted in a greater improvement in FEV₁ at 6 hours.⁶⁸ The greater FEV₁ at 30 and 60 minutes following initial albuterol doses (both 220mL [95% CI: 150 to 290]) were similar to the MCID for FEV₁ of 230mL,⁴⁷ but not associated with differences in the perception of breathlessness.⁶⁸ Albuterol resulted in a greater decrease in serum potassium levels (0.26mmol/L greater decrease at 3 hours; 95% CI: 0.16 to 0.37), higher heart rate (10 beats/min greater increase at 3 hours; 95% CI: 7.4 to 12.6) and more frequent adverse effects.⁶⁸ In studies undertaken in the ED setting, repeated administration of different reliever inhaler regimens, have shown mixed results; however, in each study budesonide-formoterol was well-tolerated and had a bronchodilator effect comparable to albuterol.^{71–73}

Pro of ICS-containing reliever therapy alone:

- i. *The clinical significance of the short-term greater bronchodilator effect of reliever albuterol compared with budesonide-formoterol when used in standard doses is uncertain.* Of concern, the greater increase in FEV₁ from repeated reliever albuterol use is associated with greater systemic beta-1 and beta-2 agonist side effects, indicating a potentially worse efficacy/safety profile when used in clinically relative doses.⁶⁸ Finally, budesonide-formoterol has a longer duration of bronchodilator action.^{69,70}

Con of ICS-containing reliever therapy alone:

- ii. *Albuterol has a greater initial bronchodilator effect and is preferred by some patients:* Albuterol has a faster onset and greater magnitude of acute bronchodilator efficacy than budesonide-formoterol when used in the comparative 200µg vs. 200–6µg dose regimens respectively.^{67,68} In one RCT comparing reliever therapy regimens, about one third of patients reported that budesonide-formoterol worked a ‘bit/a lot slower,’ one third reported that it

worked ‘about the same,’ and one third that it worked ‘a bit/a lot faster,’ when compared with the reliever inhaler they used before the study.⁷⁴ These findings indicate some patients may notice the faster onset of albuterol.

Summary: The greater reduction in severe exacerbation risk associated with budesonide-formoterol reliever use is achieved despite a lesser acute—although more prolonged—bronchodilator response. The clinical relevance of these differences in bronchodilator effects in the community setting is potentially a moot point, as patients can simply take additional doses of their reliever therapy if needed to treat persisting symptoms. Furthermore, it is salutary to remember that asthma mortality epidemics have been repeatedly associated with high-dose preparations of potent beta₂-agonists.^{9,16,17} Thus, the relatively lower beta₂-agonist exposure in reliever ICS-formoterol therapy vs. SABA therapy could offer a potentially important safety advantage if used repeatedly.

As of now, there is insufficient evidence to support the use of ICS-formoterol as a rescue medication administered by healthcare professionals in the ED or urgent care setting.^{71–73} Future studies on ICS-formoterol use in this clinical setting are needed, particularly when considering that repeated doses of ICSs in the ED have been shown to be beneficial at preventing hospitalizations.^{75,76}

4. OTHER CLINICAL AND REGULATORY CONSIDERATIONS:

- i. *Predictors of response to ICS-formoterol reliever therapy:* Whether there is a specific profile of asthma patient that preferentially responds (or does not respond) to ICS-formoterol vs. SABA reliever therapy has been assessed in two RCTs.^{48,53,77} These studies did not demonstrate evidence of an effect modification with severe exacerbation risk based on a wide range of clinical characteristics including age, sex, ethnicity, smoking status, exacerbation history, baseline SABA use, prior asthma control, lung function, blood eosinophil level, or FeNO.^{48,53,77} These findings indicate that ICS-formoterol reliever therapy is superior across the spectrum of asthma, and that clinical profiling (besides an assessment of patient preferences) is unnecessary.
- ii. *Comparing SABA and ICS-formoterol use as a marker of future exacerbation risk:* Overuse of SABAs is directly associated with an increased risk of severe exacerbations and mortality.^{80,81} In the 21-day period after a day of high reliever use the risk of a subsequent severe asthma exacerbation is decreased with high use of ICS-formoterol compared with SABA.⁸² It is likely that the continued increased use of ICS-formoterol during the 21 days after initial increased usage reduces the risk of patients progressing to a severe exacerbation, compared with continued increased SABA use. Thus, while high use of SABAs and ICS-formoterol are both markers of poor asthma symptom control, high use of SABAs as compared to ICS-formoterol is a stronger indicator of subsequent exacerbation risk.
- iii. *Simplicity of escalation to SMART:* One practical advantage of ICS-formoterol reliever therapy is the ability to easily escalate the treatment step to SMART

when indicated.^{3,6,83} This contrasts with the situation with SABA reliever therapy or ICS-albuterol reliever therapy, which calls for the use of a different maintenance ICS or ICS-LABA inhaler, according to a different schedule, when a patient's treatment is escalated.

- iv. *Limited access and higher upfront medication cost makes prescribing reliever ICS-formoterol or ICS-albuterol challenging in many locations.* Lack of access to and/or cost are significant barriers to the use of ICS-formoterol and ICS-albuterol reliever therapy in many settings.^{8,39} Budesonide-formoterol and albuterol, but not ICS-albuterol, are on the World Health Organization (WHO) Essential Medicines List.⁸⁴ Although medication availability varies dramatically between countries, the relatively high cost of ICS-containing inhalers makes their usage particularly difficult in low- and middle-income countries.^{84–86} For example, in one survey of 52 low- and middle-income countries, less than one in five public pharmacies reported that maintenance ICS inhalers were currently available.⁸⁷ Replacement of SABAs with affordable ICS-formoterol or ICS-albuterol inhalers would likely make a meaningful difference in asthma outcomes in lower-income countries where asthma is an important cause of mortality.^{84,85} Notably, as of now, budesonide-albuterol is only approved for reliever use in the US. The long-term cost of budesonide-albuterol for US patients is unknown. However, in the US, budesonide-albuterol will have new patent protections,⁵⁶ while budesonide-formoterol increasingly has generic options.⁸⁸
- v. *Regulatory barriers to ICS-formoterol reliever and ICS-albuterol reliever therapy:* Despite its approval by regulatory authorities in over 40 countries, budesonide-formoterol is not approved for use as a reliever alone by the FDA in the US, the European Medicines Agency (EMA) in the European Union (EU), or by regulatory authorities in many other countries in which ICS-formoterol maintenance and reliever therapy is approved. This regulatory situation, wherein ICS-formoterol reliever therapy alone is recommended by GINA¹ and the ERS Task Force,⁸⁹ but not approved for a reliever-alone indication in the US, EU, and numerous other countries has led to significant uncertainty for prescribers and patients,^{8,39} and limits its potential benefit for patients. Conversely, the New Zealand experience has indicated that the widespread transition to budesonide-formoterol reliever therapy-based regimens in clinical practice can be achieved with regulatory approval, and if recommended and promoted as the preferred therapeutic approach in national asthma guidelines.⁹⁰

Finally, budesonide-albuterol has been approved for use as a reliever therapy alone in adults ≥ 18 years by the FDA in the US,⁵⁶ despite the fact that the primary trial of budesonide-albuterol reliever usage only enrolled moderate-to-severe asthma patients taking concomitant ICS-based maintenance therapy and not patients on reliever therapy only.⁵⁷ To our knowledge, no other countries have approved a combination budesonide-albuterol device for reliever use. Once again, these complex regulatory situations limit the use of these therapeutic regimens globally, which likely has a detrimental effect on important patient outcomes.

Conclusion:

In adolescents and adults with mild asthma, ICS-formoterol reliever is preferred to SABA reliever therapy alone, having a better efficacy/safety profile, and in particular markedly decreasing the occurrence of severe asthma exacerbations. Data supporting the use of ICS-albuterol reliever rather than SABA reliever alone in adults with mild asthma exist but are much more limited. Entrenched medical practice, and cost and regulatory complexities are major barriers to a widespread global transition to ICS-containing reliever therapy. Furthermore, there are significant gaps in research including, most notably, limited studies of ICS-containing reliever therapy in children. We hope that explicitly describing the evidence to date will instigate future effectiveness and implementation studies of reliever-only ICS-formoterol and ICS-albuterol in mild asthma.

Financial Disclosures:

J.G.K. receives funding from the National Institute of Health and the American Lung Association and reports consulting fees outside the submitted work from Genentech and Sanofi/Regeneron. Research reported in this publication was supported by the National Heart, Lung, And Blood Institute of the National Institutes of Health under Award Number K23HL171940. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health or American Lung Association. R.B. has received personal fees from AstraZeneca, Avillion, Cipla, and Teva Pharmaceuticals outside the submitted work and is Chair of the Asthma Foundation of New Zealand adolescent and adult asthma guidelines group. The Medical Research Institute of New Zealand receives Independent Research Organization funding from the Health Research Council of New Zealand. The Medical Research Institute of New Zealand has received research funding from AstraZeneca, Genentech, and the Health Research Council of New Zealand.

Abbreviations used:

(GINA)	Global Initiative of Asthma
(SABA)	short-acting beta ₂ -agonist
(LABA)	long-acting beta-agonist
(ICS)	inhaled corticosteroid
(ED)	emergency department
(ACQ-5)	Asthma Control Questionnaire–5
(MCID)	minimal clinically important differences
(FEV ₁)	forced expiratory volume in 1 second
(FeNO)	fractional exhaled nitric oxide
(ppb)	parts per billion
(RCT)	randomized controlled trial
(DPI)	dry powder inhaler
(pMDI)	pressurized metered dose inhaler
(SMART)	single maintenance and reliever therapy

(FDA)	Food and Drug Administration
(WHO)	World Health Organization
(US)	United States
(EMA)	European Medicines Agency
(EU)	European Union

REFERENCES

1. Global Initiative for Asthma. 2023 GINA Report, Global Strategy for Asthma Management and Prevention [Internet]. 2023 [cited 2023 Oct 5]. Available from: <https://ginasthma.org/2023-gina-main-report/>
2. Reddel HK, FitzGerald JM, Bateman ED, Bacharier LB, Becker A, Brusselle G, et al. GINA 2019: a fundamental change in asthma management. *Eur Respir J*. 2019;53:1901046. [PubMed: 31249014]
3. Reddel HK, Bacharier LB, Bateman ED, Brightling CE, Brusselle GG, Buhl R, et al. Global Initiative for Asthma Strategy 2021: Executive Summary and Rationale for Key Changes. *Am J Resp Crit Care*. 2021;205:17–35.
4. O'Byrne PM, Jenkins C, Bateman ED. The paradoxes of asthma management: time for a new approach? *Eur Respir J*. 2017;50:1701103. [PubMed: 28889114]
5. Beasley R, Bird G, Harper J, Weatherall M. The further paradoxes of asthma management: time for a new approach across the spectrum of asthma severity. *Eur Respir J*. 2018;52:1800694. [PubMed: 30385601]
6. Beasley R, Bruce P, Houghton C, Hatter L. The ICS/Formoterol Reliever Therapy Regimen in Asthma: A Review. *J Allergy Clin Immunol: Pr*. 2023;11:762–772.e1.
7. Reddel HK, Bateman ED, Schatz M, Krishnan JA, Cloutier MM. A Practical Guide to Implementing SMART in asthma management. *J Allergy Clin Immunol Pract*. 2021;10:S31–8. [PubMed: 34666208]
8. Krings JG, Gerald JK, Blake KV, Krishnan JA, Reddel HK, Bacharier LB, et al. A Call for the United States to Accelerate the Implementation of Reliever Combination ICS-Formoterol Inhalers in Asthma. *Am J Resp Crit Care*. 2023;207(4):390–405.
9. Beasley R, Windom H, Pearce N, Burgess C, Crane J. Asthma mortality and inhaled beta agonist therapy. *Aust N Zealand J Med*. 1991;21:753–63. [PubMed: 1684702]
10. Aldridge RE, Hancox RJ, Robin TD, Cowan J, Winn MC, Frampton CM, et al. Effects of Terbutaline and Budesonide on Sputum Cells and Bronchial Hyperresponsiveness In Asthma. *Am J Resp Crit Care Med*. 2012;161:1459–64.
11. Cockcroft DW, Swystun VA. Functional antagonism: tolerance produced by inhaled beta 2 agonists. *Thorax*. 1996;51:1051–6. [PubMed: 8984693]
12. Mitchell EA. Is current treatment increasing asthma mortality and morbidity? *Thorax*. 1989;44:81–84. [PubMed: 2564705]
13. Wraight JM, Hancox RJ, Herbison GP, Cowan JO, Flannery EM, Taylor DR. Bronchodilator tolerance: the impact of increasing bronchoconstriction. *Eur Respir J*. 2003;21:810–5. [PubMed: 12765426]
14. Fraser PM, Speizer FE, Waters SD, Doll R, Mann NM. The circumstances preceding death from asthma in young people in 1968 to 1969. *Br J Dis chest*. 1971;65:71–84. [PubMed: 4325026]
15. Levy ML. The national review of asthma deaths: what did we learn and what needs to change? *Breathe*. 2015;11:14–24. [PubMed: 26306100]
16. Beasley R, Pearce N, Crane J, Burgess C. β agonists: What is the evidence that their use increases the risk of asthma morbidity and mortality? *J Allergy Clin Immunol*. 1999;104:S18–30. [PubMed: 10452785]
17. Taylor DR, Sears MR, Cockcroft DW. The beta-agonist controversy. *Med Clin North Am*. 1996;80:719–48. [PubMed: 8676612]

18. Ebmeier S, Thayabaran D, Braithwaite I, Bénamara C, Weatherall M, Beasley R. Trends in international asthma mortality: analysis of data from the WHO Mortality Database from 46 countries (1993–2012). *The Lancet*. 2017;390:935–45.
19. Nelson HS, Weiss ST, Bleecker ER, Yancey SW, Dorinsky PM, Group the SS. The Salmeterol Multicenter Asthma Research Trial A Comparison of Usual Pharmacotherapy for Asthma or Usual Pharmacotherapy Plus Salmeterol. *Chest*. 2006;129:15–26. [PubMed: 16424409]
20. Stempel DA, Raphiou IH, Kral KM, Yeakey AM, Emmett AH, Prazma CM, et al. Serious Asthma Events with Fluticasone plus Salmeterol versus Fluticasone Alone. *N Engl J Med*. 2016;374:1822–30. [PubMed: 26949137]
21. Suissa S, Ernst P, Benayoun S, Baltzan M, Cai B. Low-Dose Inhaled Corticosteroids and the Prevention of Death from Asthma. *N Engl J Med*. 2000;343:332–6. [PubMed: 10922423]
22. Masoli M, Holt S, Weatherall M, Shirtcliffe P, Beasley R. Inhaled corticosteroid therapy in the management of asthma in adults. In: Li JT (ed). *The Pharmacotherapy of Asthma*. New York: Taylor & Francis Group 2006; Volume 212: 83–115.
23. Barnes PJ. Current issues for establishing inhaled corticosteroids as the antiinflammatory agents of choice in asthma. *J Allergy Clin Immunol*. 1998;101:S427–33. [PubMed: 9563367]
24. Donahue JG, Weiss ST, Livingston JM, Goetsch MA, Greineder DK, Platt R. Inhaled Steroids and the Risk of Hospitalization for Asthma. *JAMA*. 1997;277:887–91. [PubMed: 9062326]
25. Reddel HK, Busse WW, Pedersen S, Tan WC, Chen YZ, Jorup C, et al. Should recommendations about starting inhaled corticosteroid treatment for mild asthma be based on symptom frequency: a post-hoc efficacy analysis of the START study. *The Lancet*. 2017;389:157–66.
26. Beasley R, Harper J, Bird G, Maijers I, Weatherall M, Pavord ID. Inhaled Corticosteroid Therapy in Adult Asthma: Time for a New Therapeutic Dose Terminology. *Am J Respir Crit Care Med*. 2019;199:1471–7. [PubMed: 30645143]
27. Boulet LP, Vervloet D, Magar Y, Foster JM. Adherence The Goal to Control Asthma. *Clin Chest Med*. 2012;33:405–17. [PubMed: 22929091]
28. Williams LK, Peterson EL, Wells K, Ahmedani BK, Kumar R, Burchard EG, et al. Quantifying the proportion of severe asthma exacerbations attributable to inhaled corticosteroid nonadherence. *J Allergy Clin Immunol*. 2011;128:1185–1191.e2. [PubMed: 22019090]
29. George M, Bilazarian A, Chung A, South K, Norful A. Reasons for inhaled corticosteroid non-adherence disclosed by African American adults during primary care visits for uncontrolled asthma. *Nurses*. 2020;5168.
30. Amin S, Soliman M, McIvor A, Cave A, Cabrera C. Understanding Patient Perspectives on Medication Adherence in Asthma: A Targeted Review of Qualitative Studies. *Patient preference adherence*. 2020;14:541–51. [PubMed: 32210541]
31. Barnes PJ. Scientific rationale for using a single inhaler for asthma control. *Eur Respir J*. 2007;29:587–95. [PubMed: 17329493]
32. Pavord ID, Beasley R, Agusti A, Anderson GP, Bel E, Brusselle G, et al. After asthma: redefining airways diseases. *The Lancet*. 2018;391:350–400.
33. Beasley R, Weatherall M, Shirtcliffe P, Hancox R, Reddel HK. Combination corticosteroid/ β -agonist inhaler as reliever therapy: A solution for intermittent and mild asthma? *J Allergy Clin Immunol*. 2014;133:39–41. [PubMed: 24369798]
34. Chapman KR, Barnes NC, Greening AP, Jones PW, Pedersen S. Single maintenance and reliever therapy (SMART) of asthma: a critical appraisal. *Thorax*. 2010;65:747. [PubMed: 20581409]
35. Chapman KR, Hinds D, Piazza P, Raheison C, Gibbs M, Greulich T, et al. Physician perspectives on the burden and management of asthma in six countries: The Global Asthma Physician Survey (GAPS). *Bmc Pulm Med*. 2017;17:153. [PubMed: 29169365]
36. DiSantostefano RL, Boudiaf N, Stempel DA, Barnes NC, Greening AP. The frequency of, and adherence to, single maintenance and reliever therapy instructions in asthma: a descriptive analysis. *npj Prim Care Respir Med*. 2016;26:16038. [PubMed: 27442488]
37. Cruz ÁA, Barile S, Nudo E, Brogelli L, Guller P, Papi A. ICS/formoterol in the management of asthma in the clinical practice of pulmonologists: an international survey on GINA strategy. *Asthma Res Pr*. 2021;7:1.

38. Chapman KR, An L, Bosnic-Anticevich S, Campomanes CM, Espinosa J, Jain P, et al. Asthma patients' and physicians' perspectives on the burden and management of asthma. *Respir Med.* 2021;186:106524. [PubMed: 34265629]
39. Krings JG, Sekhar TC, Chen V, Blake KV, Sumino K, James AS, et al. Beginning to address an implementation gap in asthma: clinicians' views of prescribing reliever ICS-formoterol inhalers and SMART in the United States. *J Allergy Clin Immunol Pract.* 2023;11:2767–2777. [PubMed: 37245736]
40. Crossingham I, Turner S, Ramakrishnan S, Fries A, Gowell M, Yasmin F, et al. Combination fixed-dose beta agonist and steroid inhaler as required for adults or children with mild asthma. *Cochrane Database Syst Rev.* 2021;2021:CD013518.
41. Sobieraj DM, Weeda ER, Nguyen E, Coleman CI, White CM, Lazarus SC, et al. Association of Inhaled Corticosteroids and Long-Acting β -Agonists as Controller and Quick Relief Therapy With Exacerbations and Symptom Control in Persistent Asthma: A Systematic Review and Meta-analysis. *Jama.* 2018;319:1485. [PubMed: 29554195]
42. Crane J, Pearce N, Burgess C, Woodman K, Robson B, Beasley R. Markers of Risk of Asthma Death or Readmission in the 12 Months Following a Hospital Admission for Asthma. *Int J Epidemiology.* 1992;21:737–44.
43. Papi A, Braithwaite I, Ebmeier S, Hancox RJ, Harrison T, Holliday M, et al. Budesonide–formoterol reliever therapy in intermittent versus mild persistent asthma. *Eur Respir J.* 2021;57:2003064. [PubMed: 33060151]
44. Rogliani P, Ritondo BL, Ora J, Cazzola M, Calzetta L. SMART and as-needed therapies in mild-to-severe asthma: a network meta-analysis. *Eur Respir J.* 2020;56:2000625. [PubMed: 32430423]
45. Juniper EF, Svensson K, Mörk AC, Ståhl E. Measurement properties and interpretation of three shortened versions of the asthma control questionnaire. *Respir Med.* 2005;99:553–8. [PubMed: 15823451]
46. Lazarinis N, Jørgensen L, Ekström T, Bjermer L, Dahlén B, Pullerits T, et al. Combination of budesonide/formoterol on demand improves asthma control by reducing exercise-induced bronchoconstriction. *Thorax.* 2014;69:130. [PubMed: 24092567]
47. Santanello N, Zhang J, Seidenberg B, Reiss T, Barber B. What are minimal important changes for asthma measures in a clinical trial? *Eur Respir J.* 1999;14:23–7. [PubMed: 10489824]
48. Beasley R, Holliday M, Reddel HK, Braithwaite I, Ebmeier S, Hancox RJ, et al. Controlled Trial of Budesonide–Formoterol as Needed for Mild Asthma. *N Engl J Med.* 2019;380:2020–30. [PubMed: 31112386]
49. Haahtela T, Tamminen K, Malmberg LP, Zetterström O, Karjalainen J, Ylä-Outinen H, et al. Formoterol as needed with or without budesonide in patients with intermittent asthma and raised NO levels in exhaled air: a SOMA study. *Eur Respir J.* 2006;28:748–55. [PubMed: 17012630]
50. Bisgaard H, Roux PL, Bjämer D, Dymek A, Vermeulen JH, Hultquist C. Budesonide/Formoterol Maintenance Plus Reliever Therapy A New Strategy in Pediatric Asthma. *Chest.* 2006;130:1733–43. [PubMed: 17166990]
51. O'Byrne PM, FitzGerald JM, Bateman ED, Barnes PJ, Zhong N, Keen C, et al. Inhaled Combined Budesonide–Formoterol as Needed in Mild Asthma. *N Engl J Med.* 2018;378:1865–76. [PubMed: 29768149]
52. Bateman ED, Reddel HK, O'Byrne PM, Barnes PJ, Zhong N, Keen C, et al. As-Needed Budesonide–Formoterol versus Maintenance Budesonide in Mild Asthma. *N Engl J Med.* 2018;378:1877–87. [PubMed: 29768147]
53. Hardy J, Baggott C, Fingleton J, Reddel HK, Hancox RJ, Harwood M, et al. Budesonide-formoterol reliever therapy versus maintenance budesonide plus terbutaline reliever therapy in adults with mild to moderate asthma (PRACTICAL): a 52-week, open-label, multicentre, superiority, randomised controlled trial. *The Lancet.* 2019;394:919–28.
54. Patel M, Pilcher J, Pritchard A, Perrin K, Travers J, Shaw D, et al. Efficacy and safety of maintenance and reliever combination budesonide–formoterol inhaler in patients with asthma at risk of severe exacerbations: a randomised controlled trial. *Lancet Respir Medicine.* 2013;1:32–42.

55. Papi A, Canonica GW, Maestrelli P, Paggiaro P, Olivieri D, Pozzi E, et al. Rescue Use of Beclomethasone and Albuterol in a Single Inhaler for Mild Asthma. *N Engl J Med*. 2007;356:2040–52. [PubMed: 17507703]
56. Food and Drug Administration. FDA approves drug combination treatment for adults with asthma [Internet]. 2023. Available from: [https://www.fda.gov/drugs/news-events-human-drugs/fda-approves-drug-combination-treatment-adults-asthma#:~:text=FDA%20has%20approved%20Airsupra%20\(albuterol,years%20of%20age%20and%20older.](https://www.fda.gov/drugs/news-events-human-drugs/fda-approves-drug-combination-treatment-adults-asthma#:~:text=FDA%20has%20approved%20Airsupra%20(albuterol,years%20of%20age%20and%20older.)
57. Papi A, Chipps BE, Beasley R Jr. RAP, Israel E, Cooper M, et al. Albuterol–Budesonide Fixed-Dose Combination Rescue Inhaler for Asthma. *N Engl J Med*. 2022;386:2071–83. [PubMed: 35569035]
58. Martinez FD, Chinchilli VM, Morgan WJ, Boehmer SJ, Lemanske RF, Mauger DT, et al. Use of beclomethasone dipropionate as rescue treatment for children with mild persistent asthma (TREXA): a randomised, double-blind, placebo-controlled trial. *The Lancet*. 2011;377:650–7.
59. Chipps BE, Israel E, Beasley R, Panettieri RA, Albers FC, Rees R, et al. Albuterol-Budesonide Pressurized Metered Dose Inhaler in Patients With Mild-to-Moderate Asthma Results of the DENALI Double-Blind Randomized Controlled Trial. *CHEST*. 2023;164:585–95. [PubMed: 37003355]
60. Mosnaim G Asthma in Adults. *N Engl J Med*. 2023;389:1023–31. [PubMed: 37703556]
61. Welsh EJ, Cates CJ. Formoterol versus short-acting beta-agonists as relief medication for adults and children with asthma. *The Cochrane database of systematic reviews*. 2010;(9):CD008418. doi:10.1002/14651858.CD008418.pub2 [PubMed: 20824877]
62. Tattersfield AE, Löfdahl CG, Postma DS, Eivindson A, Schreurs AG, Rasidakis A, et al. Comparison of formoterol and terbutaline for as-needed treatment of asthma: a randomised trial. *The Lancet*. 2001;357:257–61.
63. Rabe KF, Atienza T, Magyar P, Larsson P, Jorup C, Laloo UG. Effect of budesonide in combination with formoterol for reliever therapy in asthma exacerbations: a randomised controlled, double-blind study. *The Lancet*. 2006;368:744–53.
64. Lötvall J Pharmacological similarities and differences between β 2-agonists. *Respir Med*. 2001;95:S7–11. [PubMed: 11534896]
65. Bartow RA, Brogden RN. Formoterol. An update of its pharmacological properties and therapeutic efficacy in the management of asthma. *Drugs* 1998; 55: 303–22 [PubMed: 9506248]
66. Hanania NA, Sharafkhaneh A, Barber R, Dickey BF. β -Agonist Intrinsic Efficacy. *Am J Respir Crit Care Med*. 2002;165:1353–8. [PubMed: 12016095]
67. Kearns N, Williams M, Bruce P, Black M, Kearns C, Sparks J, et al. Single dose of budesonide/formoterol turbuhaler compared to salbutamol pMDI for speed of bronchodilator onset in asthma: a randomised cross-over trial. *Thorax*. 2023;78:745–51.
68. Kearns N, Bruce P, Williams M, Doppen M, Black M, Weatherall M, et al. Repeated dose budesonide/formoterol compared to salbutamol in adult asthma: a randomised crossover trial. *Eur Respir J*. 2022;60:2102309.
69. Noord J van, Smeets J, Raaijmakers J, Bommer A, Maesen F. Salmeterol versus formoterol in patients with moderately severe asthma: onset and duration of action. *Eur Respir J*. 1996;9:1684–8. [PubMed: 8866595]
70. Masoli M, Williams M, Weatherall M, Beasley R. The 24h duration of bronchodilator action of the budesonide/formoterol combination inhaler. *Respir Med*. 2006;100:20–5. [PubMed: 15936933]
71. Balanag VM, Yunus F, Yang PC, Jorup C. Efficacy and safety of budesonide/formoterol compared with salbutamol in the treatment of acute asthma. *Pulm Pharmacol Ther*. 2006;19:139–47 [PubMed: 16009588]
72. Boonsawat W, Charoenratanakul S, Pothirat C, Sawanyawisuth K, Seearamroongruang T, Bengtsson T, et al. Formoterol (OXIS®) Turbuhaler® as a rescue therapy compared with salbutamol pMDI plus spacer in patients with acute severe asthma. *Respir Med*. 2003;97:1067–74. [PubMed: 14509562]

73. Rubinfeld AR, Scicchitano R, Hunt A, Thompson PJ, Nooten AV, Selroos O. Formoterol Turbuhaler as reliever medication in patients with acute asthma. *Eur Respir J.* 2006;27:735–41. [PubMed: 16455838]
74. Baggott C, Reddel HK, Hardy J, Sparks J, Holliday M, Corin A, et al. Patient preferences for symptom-driven or regular preventer treatment in mild to moderate asthma: findings from the PRACTICAL study, a randomised clinical trial. *Eur Respir J.* 2020;55:1902073. [PubMed: 32029450]
75. Edmonds ML, Milan SJ, Jr CAC, Pollack CV, Rowe BH. Early use of inhaled corticosteroids in the emergency department treatment of acute asthma. *Cochrane Database Syst Rev.* 2012;2018:CD002308.
76. Kearns N, Majers I, Harper J, Beasley R, Weatherall M. Inhaled Corticosteroids in Acute Asthma: A Systemic Review and Meta-Analysis. *J Allergy Clin Immunol: Pr.* 2020;8:605–617.e6.
77. Pavord ID, Holliday M, Reddel HK, Braithwaite I, Ebmeier S, Hancox RJ, et al. Predictive value of blood eosinophils and exhaled nitric oxide in adults with mild asthma: a prespecified subgroup analysis of an open-label, parallel-group, randomised controlled trial. *Lancet Respir Med.* 2020;8:671–80. [PubMed: 32171064]
78. Lazarus SC, Krishnan JA, King TS, Lang JE, Blake KV, Covar R, et al. Mometasone or Tiotropium in Mild Asthma with a Low Sputum Eosinophil Level. *N Engl J Med.* 2019;380:2009–19. [PubMed: 31112384]
79. Deykin A, Lazarus SC, Fahy JV, Wechsler ME, Boushey HA, Chinchilli VM, et al. Sputum eosinophil counts predict asthma control after discontinuation of inhaled corticosteroids. *J Allergy Clin Immunol.* 2005;115:720–7. [PubMed: 15805990]
80. Spitzer WO, Suissa S, Ernst P, Horwitz RI, Habbick B, Cockcroft D, et al. The Use of β -Agonists and the Risk of Death and near Death from Asthma. *N Engl J Med.* 1992;326:501–6. [PubMed: 1346340]
81. Nwaru BI, Ekström M, Hasvold P, Wiklund F, Telg G, Janson C. Overuse of short-acting β_2 -agonists in asthma is associated with increased risk of exacerbation and mortality: a nationwide cohort study of the global SABINA programme. *Eur Respir J.* 2020;55:1901872. [PubMed: 31949111]
82. O'Byrne PM, FitzGerald JM, Bateman ED, Barnes PJ, Zheng J, Gustafson P, et al. Effect of a single day of increased as-needed budesonide–formoterol use on short-term risk of severe exacerbations in patients with mild asthma: a post-hoc analysis of the SYGMA 1 study. *Lancet Respir Medicine.* 2021;9:149–58.
83. Krishnan JA, Cloutier MM, Schatz M. National Asthma Education and Prevention Program 2020 Guideline Update: Where Do We Go from Here? *Am J Resp Crit Care.* 2021;203:164–7.
84. Stolbrink M, Thomson H, Hadfield RM, Ozoh OB, Nantanda R, Jayasooriya S, et al. The availability, cost, and affordability of essential medicines for asthma and COPD in low-income and middle-income countries: a systematic review. *Lancet Global Heal.* 2022;10:e1423–42.
85. Mortimer K, Reddel HK, Pitrez PM, Bateman ED. Asthma management in low and middle income countries: case for change. *Eur Respir J.* 2022;60:2103179. [PubMed: 35210321]
86. Bazargani YT, Boer A de, Leufkens HGM, Mantel-Teeuwisse AK. Essential medicines for COPD and asthma in low and middle-income countries. *Thorax.* 2014;69:1149–51. [PubMed: 24590803]
87. Babar ZUD, Lessing C, Mace C, Bissell K. The Availability, Pricing and Affordability of Three Essential Asthma Medicines in 52 Low- and Middle-Income Countries. *PharmacoEconomics.* 2013;31:1063–82. [PubMed: 24127259]
88. FDA Approves First Generic of Symbicort to Treat Asthma and COPD [Internet]. 2022 [cited 2023 Sep 24]. Available from: <https://www.fda.gov/news-events/press-announcements/fda-approves-first-generic-symbicort-treat-asthma-and-copd>
89. Papi A, Ferreira DS, Agache I, Baraldi E, Beasley R, Brusselle G, et al. European Respiratory Society Short Guidelines for the use of as-needed ICS/formoterol in mild Asthma. *Eur Respir J.* 2023;2300047. [PubMed: 37678955]
90. Hatter L, Eathorne A, Hills T, Bruce P, Houghton C, Weatherall M, et al. Patterns of Asthma Medication Use in New Zealand After Publication of National Asthma Guidelines. *J Allergy Clin Immunol: Pr.* 2023;11:2757–2764.e5.

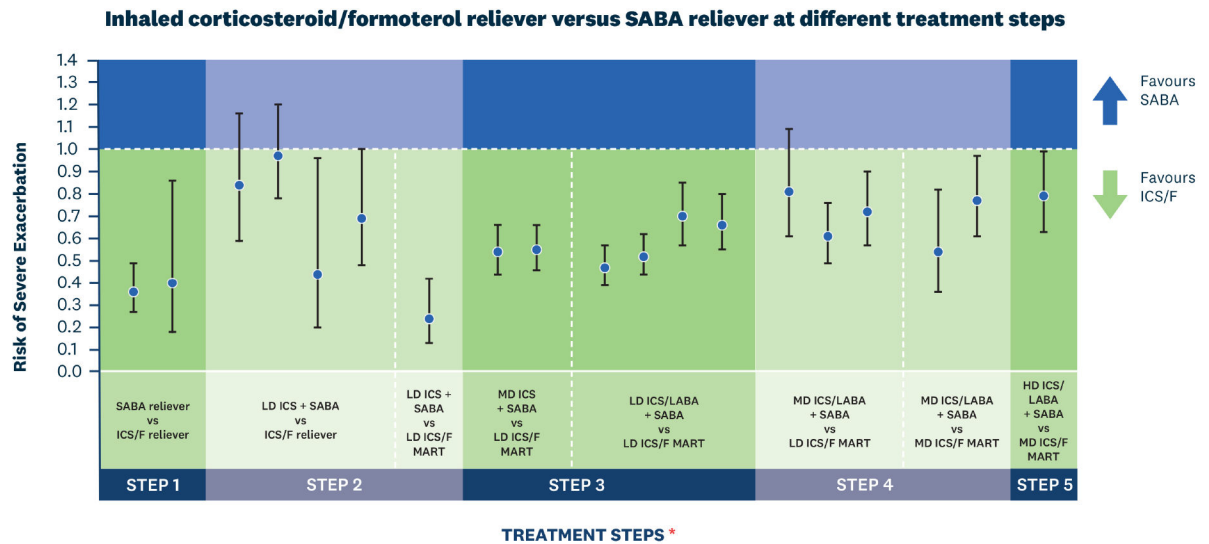
91. Beasley R, Braithwaite I, Semprini A, Kearns C, Weatherall M, Harrison TW, et al. ICS-formoterol reliever therapy stepwise treatment algorithm for adult asthma. *Eur Respir J*. 2020;55:1901407. [PubMed: 31919194]

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript



* The Treatment Steps refer to the SABA reliever-based regimen, by criteria defined by GINA 2018

Based on data summarised in Beasley R et al. ICS-formoterol reliever therapy stepwise treatment algorithm for adult asthma. Eur Respir J 2020; 55: 1901407

SABA: short-acting beta₂-agonist
ICS/F: inhaled corticosteroid/formoterol
LD: low dose; **MD:** medium dose; **HD:** high dose
MART: maintenance and reliever therapy

Figure 1:

Inhaled corticosteroid-formoterol reliever vs. SABA reliever at different treatment steps. The relative risk of severe exacerbations from randomised controlled trials of inhaled corticosteroid-formoterol vs. short-acting beta₂-agonist reliever therapy is shown for Global Initiative of Asthma Step 1–5 (although this article focuses on comparing SABA alone to ICS-containing reliever therapy in mild asthma). The risk with 95% confidence intervals is presented for each study, based on data summarized by Beasley et al.⁹¹ *The treatment steps refer to the randomised reliever-based regimen, as defined by the Global Initiative of Asthma (GINA) 2018 criteria. *Abbreviations used:* short-acting beta₂-agonist (SABA), inhaled corticosteroid-formoterol (ICS/F), low dose (LD), medium dose (MD), high dose (HD), and maintenance and reliever therapy (MART).

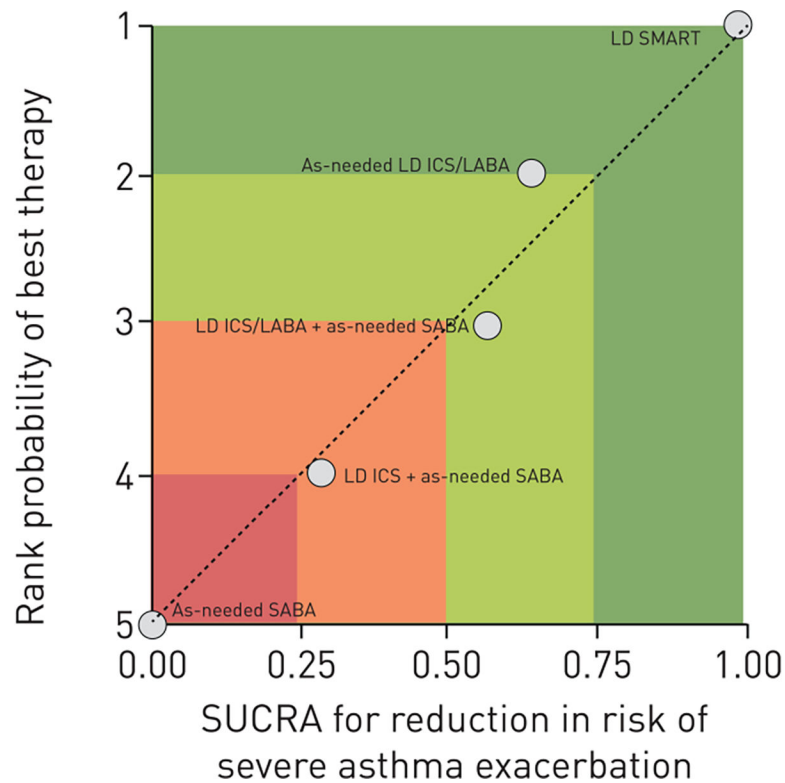
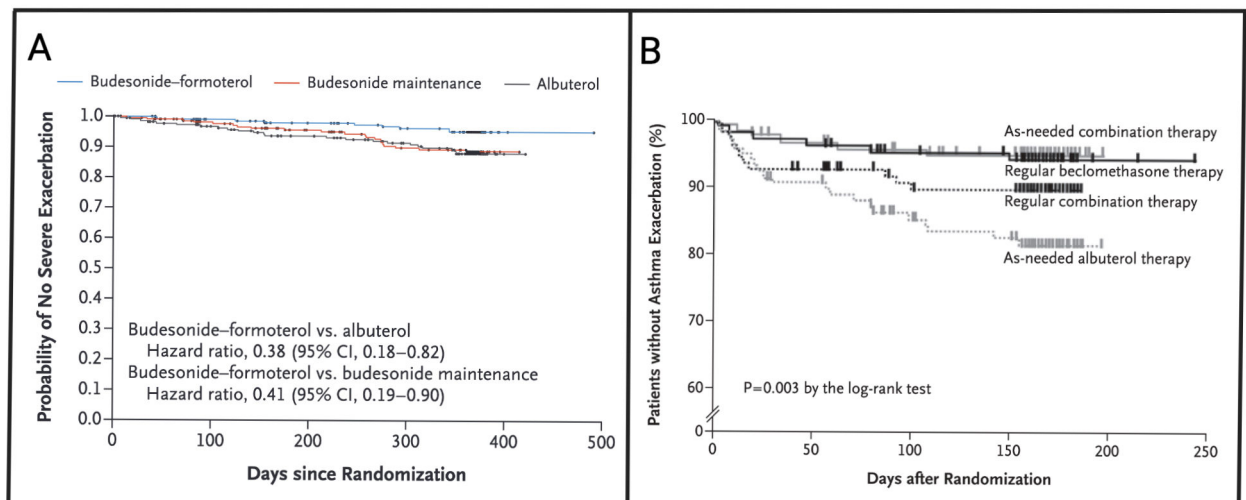


Figure 2:

Ranking plot of the efficacy of 5 therapeutic regimens in preventing the risk of severe exacerbation in mild-to-moderate asthma. Therapeutic regimens were plotted on the X-axis according to surface under the cumulative ranking curve (SUCRA), where 1 represents the treatment considered to be the best and 0 the treatment considered to be the worst. The treatments were plotted on the Y-axis according to the rank probability of best therapy, where a score of 1 is assigned to the best therapeutic strategy and 5 to the worst.

Abbreviations used: Low-dose (LD), single maintenance and reliever therapy (SMART), inhaled corticosteroid (ICS), and long-acting beta agonist (LABA). Reproduced with written permission of the ERS from Rogliani et al.⁴⁴

**Figure 3:**

Kaplan-Meier estimates of the time to first severe exacerbation based on reliever inhaler therapy regimen are shown side-by-side for inhaled corticosteroid [ICS]-formoterol and ICS-albuterol vs. short-acting beta-agonist (SABA) reliever regimens. In Panel A, results from the Novel Symbicort Turbuhaler Asthma Reliever Therapy [Novel START]) are shown for the time to first severe asthma exacerbation by randomized inhaler regimen with budesonide-formoterol reliever therapy only demarcated in blue, budesonide maintenance and albuterol reliever therapy in red, and albuterol reliever therapy only in black. In Panel B, time to first severe asthma exacerbation comparing beclomethasone-albuterol to other inhaler regimens is shown. Tick marks demarcate the first asthma exacerbation for a patient. As-needed combination therapy consisted of twice daily placebo plus reliever beclomethasone-albuterol 250–100µg in a single inhaler, regular beclomethasone therapy consisted of beclomethasone 250µg twice daily plus reliever albuterol 100µg, regular combination therapy consisted of beclomethasone-albuterol 250–100µg twice daily plus reliever albuterol 100µg, and as-needed albuterol therapy consisted of twice daily placebo plus reliever albuterol 100µg. Please note the different scales on the y axes in the Figures. Figures reproduced with written permission of the NEJM from Beasley et al⁴⁸ and Papi et al.⁵⁵

Selected comparisons of ICS-formoterol and ICS-albuterol reliever therapy alone vs. SABA reliever therapy alone in mild asthma

Table 1:

Considerations	ICS-formoterol reliever alone vs. SABA reliever alone	ICS-albuterol reliever alone vs. SABA reliever alone
Safety and efficacy:		
<i>Severe exacerbations</i>	ICS-formoterol decreases the risk of severe asthma exacerbations, and hospitalizations, ED and/or urgent care visits	Minimal data; one study demonstrated reliever beclomethasone-albuterol reduces the occurrence of asthma exacerbations, most data extrapolated from landmark trial in moderate-to-severe asthma
<i>Asthma symptoms</i>	ICS-formoterol leads to a greater improvement in asthma symptoms, but this does not meet the MCID threshold	Minimal data; one study showed ICS-albuterol improves nighttime awakenings, but uncertain clinical significance
<i>Lung function</i>	ICS-formoterol leads to a greater improvement in lung function, but this does not meet the MCID threshold	Minimal data, one study showed ICS-albuterol improves lung function, but this does not meet the MCID threshold
<i>Airway inflammation</i>	ICS-formoterol leads to a reduction in airway inflammation based on FeNO	Never studied
<i>Adverse events</i>	ICS-formoterol decreases adverse events due to fewer asthma exacerbations with no other evident safety concerns	Minimal data, one study shows no difference in adverse events
<i>Corticosteroid exposure</i>	Greater ICS, but less systemic corticosteroid exposure	Greater ICS, but less systemic corticosteroid exposure
Gaps in evidence:		
<i>Studies in adolescents and children</i>	There have been no studies of ICS-formoterol reliever therapy alone in children <12 years	There have been no studies of ICS-albuterol reliever therapy alone in patients <18 years
<i>Generalizability to different delivery devices, formulations</i>	Only studied using budesonide-formoterol DPI device	Only studied using beclomethasone-albuterol pMDI device
Bronchodilator and systemic effects^a:		
<i>Bronchodilator effects</i>	Acutely, albuterol leads to a greater initial improvement in FEV₁; ICS-formoterol leads to a greater improvement when taken repeatedly and FEV₁ is tested at hour 6	Never studied
<i>Systemic adverse effects</i>	Albuterol leads to a greater reduction in serum potassium, greater heart rate increase, and more adverse events when taken repeatedly	Never studied
Other considerations:		
<i>Role of phenotyping</i>	ICS-formoterol has been shown to be superior at preventing severe exacerbations regardless of phenotype	Never studied
<i>Escalation of therapy</i>	ICS-formoterol allows for easy escalation to SMART (using same inhaler device)	If escalating therapy and prescribing maintenance therapy, patient needs to fill 2 different inhalers
<i>Cost</i>	ICS-formoterol is not cost accessible for many patients, particularly in low- and middle-income countries; generic budesonide-formoterol options are entering the marketplace in the US and globally	Few countries have access to ICS-albuterol, cost accessibility of ICS-albuterol remains unknown in the US
<i>Regulatory considerations</i>	ICS-formoterol approved as a reliever therapy in >40 countries; however, not approved as a reliever inhaler alone in the US or EU	ICS-albuterol approved by the FDA as a reliever inhaler despite relative paucity of data on its use in mild asthma; unavailable in other countries

Table 2:

ICS-formoterol reliever therapy vs. SABA reliever therapy: Efficacy and safety

Outcome (with associated citation)	Result	Certainty of evidence #
Severe exacerbations ⁴⁰	OR 0.45, 95% CI 0.34 to 0.60	High
Hospital admissions/ED or urgent care visits ⁴⁰	OR 0.35, 95% CI 0.20 to 0.60	Low
ACQ-5 ⁴⁰	Mean difference: -0.15, 95% CI -0.20 to -0.10	Moderate
Pre-bronchodilator FEV ₁ ⁴⁰	Mean difference 64mL, 95% CI 27 to 101	Moderate
FeNO ⁴⁸	Ratio of geometric means 0.83, 95% CI 0.75 to 0.91	Low
Adverse events ⁴⁰	OR 0.82, 95% CI 0.71 to 0.95	Moderate
Inhaled corticosteroid dose (budesonide ug/day) ⁴⁰	Mean Difference: 76.5 gg, 95% CI 69.4 to 83.6	Moderate
Systemic corticosteroid dose (prednisone mg/year) ⁴⁰	Mean difference -9.9 mg 95% CI 19.4 to 0.4	Low